BRIAN JARRETT

Solutions Architecture & Technical Leadership | Cloud, Autonomous & Platform Infrastructure

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PROFESSIONAL SUMMARY

Hands-on Technical Leader and Solutions Architect with a 30-year systems engineering foundation spanning bare-metal compute, private cloud infrastructure, DevSecOps automation, and federal compliance. A practical problem-solver who understands systems from the ground up, verifies data directly at the source, and designs automated tools, data platforms, and ETL pipelines that eliminate manual error and resolve complex operational challenges. Proven track record of entering unfamiliar technical ecosystems, rapidly deconstructing complex architectures, and translating intricate enterprise and federal requirements (FedRAMP / STIGs) into reliable, maintainable platforms. Demonstrated leadership directing enterprise zero-day incident response across 6,000+ nodes to defend FedRAMP ATO with the HHS PMO, building custom vulnerability data platforms and executive dashboards, and serving as the technical bridge across multidisciplinary software, networking, and platform teams.

SKILLS

Data & Systems Engineering

- **Python (Pandas, SQLAlchemy):** Large-scale data manipulation, automated reporting, ETL pipelines, and API integrations.
- **ETL Pipelines:** Architecting high-integrity data pipelines, database migrations, and schema validation.
- **PostgreSQL, MySQL:** Relational database operations, replication, schema migration, and performance query tuning.
- **Validation & QA Testing:** Rigorous test case design, orchestration testing, and end-to-end verification methodologies.
- **Hardware-in-the-Loop (HIL):** Validating edge software and control loops against physical sensors, actuators, and motor controllers before full assembly.
- **AI & RAG Orchestration:** Local multi-agent LLM orchestration (LiteLLM, Hermes), vector databases (Qdrant), and graph analytics.

Systems & Distributed Scale

- **GPU Infrastructure & Operator:** PCIe VFIO driver binding, IOMMU isolation, NVIDIA GPU Operator, CDI, time-slicing multi-tenancy, and DCGM telemetry.
- **OpenShift & Kubernetes Platforms:** Declarative platform engineering, RHCOS MachineConfigs, custom SCCs, SNO deployment automation, and KServe overlays.
- **AI Serving & SRE Runbooks:** High-throughput model serving (vLLM, continuous batching, PagedAttention), Prometheus telemetry, PCIe GPU recovery runbooks, and post-mortem RCAs.
- **Cisco UCS Hardware:** Enterprise bare-metal compute and fabric interconnect administration.
- **OpenStack:** Public/private cloud orchestration, scheduler tuning, and infrastructure verification.
- **vSphere / ESXi / Proxmox:** High-density enterprise virtualization, clustering, and VM lifecycle automation.

- **Ceph Block Storage (RBD):** High-availability distributed storage clusters, snapshot management, and performance benchmarking.
- **HA Clustering:** Designing failover systems and fault-tolerant identity/compute clusters.
- **Low-SWaP Edge Devices:** Deploying and stress-testing containerized workloads across low-power ARM64 single-board computers and air-gapped testbeds.
- **Linux (RHEL/Ubuntu):** Kernel tuning, package management, lights-out management (LOM), and shell automation.

DevOps, Security & Compliance

- **Ansible, SaltStack:** Automated configuration management, fleet audits, and rapid node provisioning.
- **CI/CD (GitHub Actions, Jenkins):** Automated test execution, artifact generation, and deployment pipelines.
- **FreeIPA, RBAC/HBAC:** Centralized authentication and role-based access control across thousands of heterogenous nodes.
- **FedRAMP & STIG Compliance:** Automated compliance-as-code extraction, auditing, and continuous security baseline enforcement.
- **Qualys Security Intelligence:** Automated vulnerability scanning, compliance metric extraction, and remediation tracking.
- **Network Engineering & Firewalls:** Cisco routing/switching, IPsec VPNs, VLANs, Ruckus wireless bridges, and VoIP architecture.

Leadership, Mentoring & Methodology

- **Solutions Architecture & Integration:** Translating enterprise and federal requirements into reliable data platforms, automated infrastructure, and operational workflows.
- **Technical Mentorship:** Guiding multi-tier engineering teams, methodical troubleshooting, and stakeholder communication.
- **Robotics & STEM Mentoring:** Mentoring student teams in real-time embedded systems, Python control loops, sensor fusion, computer vision (AprilTags and ML object detection), and HIL testing.
- **Advanced Diagnostics & Monitoring:** Low-level system diagnostics, proactive performance telemetry, and elimination of systemic issues.
- **Rapid Adaptation / SME:** Proven track record of mastering unfamiliar architectures quickly and becoming an organizational SME.

PROFESSIONAL EXPERIENCE

Cisco Systems | Cloud Engineering Technical Leader

AUG 2019 – JULY 2026 (REMOTE)

Managed FedRAMP Moderate operations, backend data systems, and compliance automation for the Webex for Government platform, providing technical leadership across corporate Webex and SecOps systems.

- **Federal ATO Governance & Enterprise Log4j Response (HHS PMO):** Played an integral technical role in the enterprise Log4j response affecting 6,000+ operational bare-metal and cloud systems under the HHS PMO’s FedRAMP ATO governance. Built custom Ansible and Python deep-inspection tooling to locate nested/embedded Java libraries missed by commercial scanners, delivered actionable findings to engineering teams and directors, verified remediation, and helped compile the evidence packages used in PMO reporting that defended the platform’s ATO.
- **Vulnerability Data Platform & Fleet UI Architecture (FastAPI, PostgreSQL, Tableau):** Designed and maintained an internal FastAPI + PostgreSQL vulnerability portal (fed by automated Jira ETL) used by multiple engineering teams to triage and plan remediation, and by leadership to track and report fleet-wide status

across the Webex FedRAMP environment. Resolved operational alert fatigue caused by fragmented 1:1 server-to-ticket reporting by condensing issues into actionable fleet views. Extended the platform with Tableau dashboards segmented by team and SLA severity, empowering leadership to drive continuous SLA adherence.

- **Cross-Discipline Technical Translation & Federal Deviation Authority:** Served as the technical bridge between software developers, storage engineers, and networking teams, demystifying abstract CVEs, kernel vulnerabilities, and DISA STIG benchmarks into actionable code and configuration remediation paths tailored to each stack. Researched root causes, evaluated production risk, and authored formal technical compliance deviation and risk justification packages submitted to the federal PMO for false positives and acceptable operational risks.
- **Enterprise Modernization & Qualys Grouping Taxonomy (ServiceNow Migration):** Served as lead technical consultant for the enterprise-wide modernization effort to retire the legacy Jira tracking system in favor of an automated ServiceNow integration. Partnered with the ServiceNow ETL engineering team to architect asset tagging taxonomies, dynamic grouping structures, and metadata classification rules inside Qualys, ensuring vulnerability and compliance findings were accurately classified and dynamically routed to the appropriate engineering owners.
- **DevSecOps & Compliance Automation (STIG XCCDF/XML):** Built custom automation tooling in Python and REST APIs to ingest and parse raw DISA STIG XCCDF (XML) benchmark data, mapping running Linux server configurations directly to federal controls. Automated the continuous generation of audit-ready STIG checklists, slashing hundreds of engineering hours of manual compliance preparation and ensuring audit readiness across all Webex for Government footprints.
- **Engineering Mentorship & Technical Guidance:** Mentored junior and mid-level engineers (up to three concurrent early in the role; later one-at-a-time onboarding) on Linux diagnostics, root-cause troubleshooting, and operational practices. Supported the hiring process through interviewing and structured onboarding of new team members across distributed remote teams, while providing ongoing technical guidance on complex remediation under federal SLAs.
- **Inventory Analytics & Infrastructure Operations:** Leveraged Pandas for large-scale data manipulation, merging disparate inventory reports and datasets into actionable operational insights across Cisco UCS bare-metal hardware and cloud compute. Transformed manual Excel-based tracking into automated trend analysis for monthly executive reporting.
- **Skills:** Python (FastAPI, Pandas, SQLAlchemy), PostgreSQL, Jira API ETL, Qualys API, Tableau, DISA STIGs (XCCDF/XML), Federal ATO Governance (HHS PMO), Ansible, RHEL/Ubuntu, Cisco UCS, FedRAMP Moderate.

Cisco Systems | Cloud Systems Engineer & Lead Compute Engineer

JAN 2017 – AUG 2019 (REMOTE)

Served as Lead Compute Engineer managing compute, identity, and bastion infrastructure across Webex Commercial and FedRAMP Moderate footprints. Collaborated closely with Networking, Storage, and Security Operations Center (SOC) teams to guarantee zero-downtime availability, fine-grained access control, and continuous security hardening.

- **Enterprise Identity Architecture & SOC Collaboration (FreeIPA/HBAC):** Architected and deployed a High-Availability (HA) FreeIPA cluster in OpenStack using Ansible to replace a legacy, high-overhead 389 Directory Server environment. Partnered directly with the Security Operations Center (SOC) to architect and define Host-Based Access Control (HBAC) rule sets—introducing net-new zero-trust access control capabilities—while guiding disparate application engineering teams on service account permissions and access policies across 2,000+ Linux/Windows servers and network appliances.
- **Lead Compute Engineering & Virtualization Operations:** Managed large-scale compute services and bastion host environments using VMware vSphere on Cisco UCS bare-metal hardware across commercial and FedRAMP data centers. Served as lead compute engineer resolving complex virtualization performance

bottlenecks, ensuring high availability, maintaining strict adherence to federal uptime SLAs, and supporting SAN storage fabric operations including Cisco MDS Fibre Channel switches and zoning.

- **Compute Security & Vulnerability Remediation:** Led security patch remediation and hardening efforts for compute services and bastion hosts in highly regulated federal environments. Researched and validated system security patches, advised engineering teams on remediation strategies, and managed Qualys vulnerability scanners to ensure continuous FedRAMP compliance.
- **Secure Access & Bastion Infrastructure (Bastions, LDAP, Active Directory):** Architected and maintained the secure access infrastructure across Webex commercial and FedRAMP footprints, managing hardened Linux bastion hosts, Active Directory domain controllers, and LDAP directory services to enforce zero-trust access control and secure remote administrative access across 2,000+ systems.
- **Skills:** Ansible, FreeIPA (HA/HBAC), OpenStack, VMware vSphere, Cisco UCS, RHEL/CentOS, Windows Server, RBAC/HBAC, Qualys, FedRAMP Compliance.

Select Group @ Cisco | Cloud DevOps/QA Engineer (Contract)

FEB 2016 – JAN 2017 (REMOTE)

Provided specialized DevOps and QA engineering support for Cisco's OpenStack-based Public Cloud (InterCloud Services) running an OpenStack-on-OpenStack environment on Cisco UCS bare metal. Focused on Nova scheduler performance optimization, automated configuration management, and stability engineering for large-scale cloud deployments.

- **Performance Optimization:** Identified and removed redundant **Nova Scheduler** filters, reducing instance scheduling time for 700 nodes from 20 seconds to 5 seconds—a **75% improvement** in orchestration speed.
- **Bare-Metal PXE Provisioning & Automation at Scale:** Engineered automated bare-metal deployment and provisioning workflows using **PXE boot** and **Ansible** playbooks across hundreds of production servers within Cisco's InterCloud Services. Automated ongoing configuration management and system maintenance via **lights-out management (LOM)** interfaces and scheduled orchestration tasks, eliminating manual intervention and ensuring consistent, repeatable hardware baselines across large-scale cloud footprints.
- **Cloud Infrastructure Evaluation:** Stood up **RHUI (Red Hat Update Infrastructure)** and **Proxmox** clusters for **Proof-of-Concept (PoC)** assessments to determine viability for integration with Cisco InterCloud Services.
- **Stability Engineering:** Diagnosed and resolved a critical race condition involving cloudbase-init on Windows 2008/2012 images within **OpenStack**, eliminating intermittent reboots during initial instance setup.
- **Cloud QA & Orchestration:** Managed global test cycles for Compute and Heat components during major cloud upgrades. Developed **Python** and **Bash** scripts coupled with Ansible to perform comprehensive orchestration tests.
- **Skills:** OpenStack (Nova, Heat, Neutron, Glance), Ansible, Python, Bash, RHUI, Proxmox, Linux/Windows Administration, Performance Tuning, Cloud QA.

Australian Laboratory Services | Senior Systems & Storage Administrator

SEP 2014 – JAN 2016 (GRAND JUNCTION, CO)

Managed, modernized, and optimized enterprise storage, virtualization, database, and network infrastructure. Engineered high-availability distributed storage clusters, automated server provisioning, and implemented comprehensive monitoring to ensure maximum uptime, data integrity, and network performance.

- **Distributed Storage & Virtualization:** Engineered an Ubuntu-based **Ceph cluster** for **VMware ESXi** hosts, benchmarking iSCSI vs. RBD performance to optimize block storage access. Leveraged **Ceph RBD snapshots** to manage automated database schema testing, enabling a major production database upgrade with zero errors.

- **Infrastructure Automation:** Implemented **SaltStack** to manage Linux virtual environments and automate the configuration and rapid provisioning of new **Ceph** storage nodes and virtual machine resources.
- **Database & Data Integrity:** Wrote custom **Python (SQLAlchemy)** verification scripts to check for missing SQL foreign keys and ensure data integrity during major migrations. Configured replication and backups of production **MySQL** databases using **LVM snapshots** and **rsync**.
- **System Stability & Monitoring:** Installed **Observium** for performance telemetry and integrated proactive checks into **Nagios** to monitor production systems, decreasing service restarts from weekly to quarterly.
- **Network Hardware Engineering:** Troubleshot and resolved a complex routing issue where **Asterisk** VoIP traffic was throttled by backup data tunnels; replaced overloaded gateway hardware, doubling available network bandwidth.
- **Skills:** Ubuntu/CentOS, VMware ESXi, Ceph RBD, iSCSI, SaltStack, Python (SQLAlchemy), MySQL, Nagios, Observium, Asterisk VoIP, Cisco/HP Routers & Switches, Sophos UTM, VLANs, IPsec VPN.

Garfield County School District 16 | Senior Systems/Network Administrator
OCT 2002 – JUN 2014 (PARACHUTE, CO)

Managed and modernized the district-wide enterprise network, systems, and core educational platform infrastructure. Led high-impact IT modernization projects, automated administrative processes, and engineered cost-effective systems that dramatically reduced operational costs and IT support overhead.

- **Systems Integration & Automation:** Developed a custom **Flask (SQLAlchemy)** web application that synchronized the centralized Student Information System (SIS) with **Active Directory** and **Google Apps**. Built self-service password reset and status tools for teachers, resulting in a **97% reduction in recurring support calls** to the helpdesk.
- **Enterprise Modernization & Development:** Spearheaded the replacement of a legacy, fragmented student information system with a web-based, centralized platform, eliminating manual data syncing between schools. Streamlined state compliance reporting by developing custom **PHP** generation scripts interfacing with the **PostgreSQL** database.
- **Infrastructure & Systems Virtualization:** Designed and administered virtualized host environments using **Citrix XenServer** and **VirtualBox** to run critical Windows Server 2008 and Ubuntu Linux servers. Supervised low-voltage cabling and hardware installation contractors for new facilities to ensure strict specification compliance.
- **Network Engineering:** Engineered a high-speed WAN connection between district buildings using **Ruckus wireless bridges**, increasing bandwidth by **6,500%** (1.5Mbps to 100+ Mbps) with zero ongoing monthly service costs. Managed core networking infrastructure including **Cisco Routers, Switches, PIX Firewalls**, and **Smoothwall** web filters.
- **Unified Communications & End-User Compute:** Deployed **Cisco CallManager** and **Cisco Unity** voice systems district-wide. Architected a centralized desktop computing solution using **Windows Terminal Servers** and **HP Thin Clients** for 6 computer labs, saving the district over **$60,000** in capital expenditure and simplifying end-user support.
- **Skills:** Python, Flask, SQLAlchemy, PHP, PostgreSQL, Ubuntu, Windows Terminal Server, Windows AD, Citrix XenServer, VirtualBox, Cisco CallManager/Unity, Cisco Routers & Switches, PIX, Ruckus Wireless Bridges/APs, HP Thin Clients, Google Apps, Smoothwall, Active Directory.

Foundational Experience | Systems Engineering, Software QA & Avionics Foundations

1991 – 2002 (VARIOUS)

Established a lifelong technical foundation by translating a low-level understanding of hardware, networking, and software into rigorous testing and validation methodologies. This ground-up engineering approach ensures comprehensive diagnostics and root-cause analysis at every layer of the stack.

- **Sales Engineering & Systems Integration:** Tested **ImageCast** software integrations with systems management suites including **SMS**, **CA Unicenter**, and **Desktop DNA**. Defined customer software requirements, submitted feature change requests, and conducted technical product demonstrations and training for enterprise clients such as **HP**, **Target**, and **NEC**. Helped answer technical questions at trade shows.
- **Technical Services & Test Lab Management:** Merged and managed QA testing with the technical support department, hiring and mentoring technicians on networking topologies and structured testing methodologies. Designed test cases, executed testing plans, and managed the lab's **Windows** and **Linux** servers and test networks across diverse hardware. Redesigned, built out, and managed the company's network, labs, and server infrastructure following a corporate split.
- **Software QA & Testing Leadership:** Promoted from contractor to QA Lead, managing all testing operations for **ImageCast** and **DrivePro** products. Structured software testing workflows by implementing the company's first bug-tracking system, drafting comprehensive test plans, and maintaining lab hardware and network infrastructure.
- **Training & Team Optimization:** Boosted support staff effectiveness and QA productivity by rotating tech support personnel through active lab testing scenarios. Trained technicians on networking topologies, testing methodologies, and product architecture to improve time-to-resolution for support issues.
- **Hardware Assembly & Diagnostics:** Built custom computers, performed component-level hardware and peripheral repairs, and conducted software troubleshooting. Developed quotes for client builds and mastered motherboard/component compatibility.
- **Airborne Radar & Electronics Engineering (U.S. Air Force):** Served as an Airborne Warning and Control Radar Journeyman (AWACS), repairing high-power microwave systems, high-voltage circuits, and analog/digital electronics.
- **Skills:** Analog & Digital Electronics, Circuit Analysis, System Diagnostics (Binary/Hex), PC Architecture, Disk Cloning (ImageCast, DrivePro), Software QA & Test Automation, Bug-Tracking Systems, Network Design, Windows/Linux Lab Administration, Technical Training, Requirements Engineering, Systems Management (SMS, CA Unicenter, Desktop DNA).

INNOVATION LAB & COMMUNITY LEADERSHIP

The Innovation Lab (Personal R&D)

CONTINUOUS HANDS-ON RESEARCH AND PROTOTYPING ENVIRONMENT FOCUSING ON CLOUD-NATIVE INFRASTRUCTURE, AIR-GAPPED DELIVERY PLATFORMS, SOVEREIGN AI ORCHESTRATION, AND DATA PIPELINE ENGINEERING ON DEDICATED WORKSTATION INFRASTRUCTURE.

- **OpenShift & Kubernetes AI Infrastructure (openshift-ai-infrastructure):** Hands-on research and applied SRE platform for enterprise AI stacks, bridging bare-metal hardware and Kubernetes. Engineered host PCIe VFIO passthrough and IOMMU isolation for NVIDIA GPU acceleration, deployed the NVIDIA GPU Operator with CRI-O Container Device Interface (CDI) and time-slicing multi-tenancy, and orchestrated production OpenAI-compatible vLLM inference (PagedAttention, continuous batching) alongside KServe/RHOAI declarative overlays. Formulated Day-2 SRE operational runbooks (PCIe bus reset, node drain), DCGM/Prometheus alerting for KV-cache and P95 latency, and incident post-mortems.

- **Single-Node OpenShift Platform Substrate (openshift-sno-lab):** Architected an automated, declarative infrastructure substrate deploying Single Node OpenShift (SNO 4.16+) via OpenTofu and KVM/libvirt, establishing strict separation of concerns between infrastructure provisioning and AI workloads. Implemented a flat peer-bastion topology (split-DNS, HAProxy ingress) and dedicated isolated bridge (virbr-ocp) ensuring zero host collision; tuned NVMe PCIe storage pools (cache=none, io=threads) to guarantee sub-10ms etcd fsync latency. Enforced zero-trust SSH with native OpenTofu Ed25519 deterministic key pinning and authored declarative RHCOS MachineConfigs for boot-time 1GB HugePages and kernel IOMMU passthrough.
- **Tactical Edge Telemetry Gateway (tactical-edge-gateway):** Architected an air-gapped telemetry translation and operator situational awareness gateway bridging autonomous robotics protocols (**MAVLink and ROS2/DDS**) to military C2 (**Cursor on Target / TAK Server**) over contested DDIL links. Authored Architecture Decision Records (ADRs) and threat models, packaging services as zero-dependency **Zarf OCI artifacts** with continuous **Lula (OSCAL)** compliance validation.
- **Tactical Edge Software-Defined Networking (tactical-edge-sdn):** Engineered a cloud-native software-defined networking reference architecture modernizing edge and shipboard tactical routing. Features an autonomous Python/C++ SD-WAN controller that evaluates latency and packet loss across simulated radio links, mutating Linux Netlink kernel routes dynamically with zero dropped sessions, packaged with containerized FRRouting and WireGuard as Zarf packages.
- **Air-Gapped Delivery & cATO Prototyping (Defense Unicorns Zarf/UDS/Lula):** Evaluated declarative air-gapped platform delivery using **Defense Unicorns' Zarf** and **UDS bundles**. Integrated **Lula** with **OSCAL** schemas to validate live Kubernetes resources against NIST SP 800-53 security controls, automating the generation of compliance artifacts and System Security Plans (SSPs).
- **Immutable Kubernetes Infrastructure (Talos Linux & eBPF Cilium):** Engineered an immutable, declarative Kubernetes control plane using **Talos Linux** (API-managed, read-only rootfs, zero-SSH attack surface). Configured **eBPF-powered Cilium** for high-performance socket routing, L2 BGP load balancing, and observability.
- **Automated Sandbox Substrates & Deterministic Security (OpenTofu & Libvirt):** Built automated multi-tier sandbox topologies replicating isolated network zones and jump hosts using **OpenTofu** on KVM/Libvirt. Authored architectural standards for deterministic SSH host key generation to eliminate security anti-patterns in ephemeral testing.
- **Multi-Agent AI Orchestration & MLOps:** Engineered a private, server-grade AI orchestration stack using **LiteLLM** and **Hermes** on a dedicated T5600 workstation (128GB RAM/RTX 3060).
- **Knowledge Retrieval & Vector Search:** Implemented a **RAG (Retrieval-Augmented Generation)** pipeline utilizing **Qdrant** as a vector database to enable natural language querying across a multi-terabyte technical knowledge base.
- **Graph Analytics & Entity Mapping:** Utilizing **Graphifyy** to engineer complex network graphs for relationship mapping, entity extraction, and knowledge discovery.
- **Statistical Research & Graph Clustering:** Exploring advanced community detection and data clustering using the **Leiden algorithm** to derive structural insights from unstructured datasets.

FIRST Robotics Mentorship

ROLE: LEAD TECHNICAL SYSTEMS MENTOR | **TIMELINE:** ONGOING

Mentoring high school engineering students in mechanical/electrical design, real-time embedded programming, and competitive robotics systems. Instilling rigorous diagnostic and troubleshooting methodologies under high-pressure tournament conditions.

- **Real-Time Embedded Systems, Controls & HIL Subsystem Testing:** Mentoring student teams in Python robot control architectures, sensor fusion, CAN bus communications, and camera-based computer vision (AprilTags and ML object detection). Executing hardware-in-the-loop (HIL) testing on physical subsystems (sensors, actuators, and motor controllers) before full robot assembly.

- **Methodical Diagnostic Methodology:** Training students in systematic root-cause troubleshooting for electrical, software, and mechanical subsystems to rapidly diagnose and resolve failure points during active competition.
- **Engineering Collaboration & Leadership:** Guiding students through the iterative engineering design process, technical documentation, design reviews, and collaborative problem solving.

EDUCATION & CERTIFICATIONS

- **Linux Academy / A Cloud Guru:** AWS & Red Hat Enterprise Linux Curriculum
- **Regis University:** C Programming & Unix Systems Courses
- **Lincoln Christian University:** Music Theory & Composition
- **Colorado School of Mines:** 2 Years – Electrical Engineering & Applied Physics
- **Community College of the Air Force:** Avionics/Electronics Technician – RADAR Specialist (AWACS)
- **On the Job Training – Tinker AFB, OK:** AWACS RADAR and IFF Systems Journeyman – Awarded Senior Airman Below-the-Zone
- **Air Force Technical Training School – Keesler AFB, MS:** Electronics and AWACS RADAR Courses – Honor Graduate (99% Academic Test Average)

For a list of legacy certifications, see linkedin.com/in/bjarrett/details/certifications/